

VUFIX: Designing Frictionless Hostel Maintenance

A dual-role progressive UX ecosystem connecting 1,200+ residential students with campus facilities administration.



Student Presenter

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Rubrics Alignment (40M)

- Usability Testing & Analysis (25%)
- Peer Benchmarking (25%)
- 5-Min Presentation (25%)
- Behance Case Study (25%)



Architecture & System

Single-Page Application, LocalStorage Reactive Store, Live Stepper, Two-Way Warden Escalation Engine.

The "Black Hole" of Dormitory Maintenance

Paper logbooks and fragmented WhatsApp groups caused systemic failure across 4 residential blocks.



Zero Status Transparency

Students file physical complaints in hostel registers with no tracking, leading to endless anxiety and repeated visits to the warden office.



Misrouted Technician Triage

Wardens lacked structured categorization, routinely dispatching electrical technicians to plumbing emergencies, creating 48hr+ resolution lags.



Phantom Resolutions

Staff marked tickets "Done" on paper clipboards without resident verification, forcing students to re-register issues when repairs failed.

Dual-Role Synchronized System Architecture

Purpose-built interfaces for both residential Students and the Hostel Warden Administration.



Student Portal (Aryaman Saboo)

- **Pre-Authenticated Identity:** Automatically binds Hostel A, Block B3, Room B3-304.
- **Guided 3-Step Wizard:** Category tiles → Issue details → Confirmation.
- **Live Stepper:** Transparent milestone progress with assigned technician & ETA.
- **Two-Way Verification:** Explicit "Yes, Fixed" vs "No, Still Broken" loop.



Warden Portal (Warden Rinu Babu)

- **High-Density Triage Dashboard:** 5 real-time KPI stat cards with 1-click filters.
- **Urgent Escalation Banner:** High-priority visual alert for delayed issues.
- **Specialist Staff Dispatch:** Auto-matching trade roster (Plumbing, Electrical, IT).
- **Live Student Timeline Sync:** Push internal warden notes directly to student app.

Usability Testing & Broken Nielsen Heuristics

Evaluated across 5 real campus users (Aryaman, Ananya, Kabir, Priya, and Warden Asst. Dilip Rao).

Heuristic #2: Match System & Real World

Friction: Users hesitated between "Electrical" vs "Furniture" when reporting desk lamp socket repairs.

Fix: Enriched categories with contextual sub-labels (Electrical: Fan, Light, Socket, Geyser).

Heuristic #5: Error Prevention

Friction: Freshmen clicked "Escalate to Warden" 10 minutes after submitting, assuming it was a fast-track button.

Fix: Implemented dynamic SLA timer gating ("Eligible in 18 hrs") and mandatory reason prompt.

Heuristic #6: Recognition Rather Than Recall

Friction: Warden assistant struggled to remember staff specialties from plain employee names.

Fix: Added trade specialty badges & auto-recommendation matching in staff dispatch dropdown.

Heuristic #3: User Control & Freedom

Friction: Students hesitated to mark "No, Still Broken" fearing loss of prior repair records.

Fix: Added explicit modal assuring students that reopening keeps full timeline history intact.

System Architecture vs Unpredictable Human Behavior

Stress-testing the reactive state layer against erratic human interaction patterns.



Rapid Double-Clicking

Behavior: Users rapidly mashed submit button on slow connections.

Fix: State-level debouncing and submission lock flag preventing duplicate database writes.



Multi-Tab Desync

Behavior: Student escalated on Tab 1 while Warden resolved on Tab 2.

Fix: Window `storage` event broadcast listeners refreshing in-memory state automatically.



Malicious Input Payloads

Behavior: Users pasted raw HTML/scripts into issue descriptions.

Fix: Strict `textContent` node rendering and CSS break-word bounds preventing DOM injection.

Comparative Analysis Across 3 Classmates

Analyzing alternative Information Architectures and UI logic paradigms with fellow students.

CLASSMATE / PROJECT	THEIR UI ARCHITECTURE	VUFIX STRUCTURAL CHOICE	KEY LEARNING ADOPTED
Kunal / Ganesh Hostel Maintenance	Monolithic single-scroll long form + WhatsApp ping	Guided 3-Step Wizard with visual category selector cards	Added direct 1-tap `tel:` phone shortcut to assigned technician profile for urgent call-backs.
Prasad IT Helpdesk & SLA	Strict ITIL 8-state model with automated countdown timers	Streamlined 5-milestone visual stepper with student escalation	Incorporated explicit Resolution ETA timestamps set by Warden during technician assignment.
Neermay Campus Facilities	Dorm door QR scan with 48hr auto-closure policy	Pre-bound authenticated profiles with active dual confirmation	Refined visual glassmorphic vertical stepper cards and timestamp badges for status clarity.

Usability Leap (SUS 64.5 → 89.0) & Viva Voce

Measurable empirical validation and complete readiness for examination evaluation.

89.0

System Usability Scale (SUS)

Jumped from 64.5 (Grade D) to 89.0 (Grade A+ / 97th percentile).

96%

Task Completion Rate

Increased from 76% baseline. Zero abandoned complaint filings.

38s

Average Filing Time

Slashed by 69% from 124s. Frictionless dormitory maintenance.

✔ ALL 4 RUBRICS FULFILLED: Usability Testing (25%) · Peer Benchmarking (25%) · 5–Min Presentation (25%) · Behance Case Study (25%)

Ready for Viva Voce Q&A